

ASSIGNMENT 2

• MODULE 14

1. Given the following pseudo-code for a function that calculates delivery charges based on order amount, write white-box test cases to achieve 100% statement coverage.

function calculateDeliveryCharge(amount):

if amount >= 500:

return 0

else:

return 50

ANS.

achieve **100% statement coverage**, every statement (return 0 and return 50) must be executed at least once.

TC ID	Input (Amount)	Expected Output	Statements Covered
TC1	600	0	if branch → return 0
TC2	400	50	else branch → return 50

Coverage Achieved: 100% Statement Coverage

2. Write white-box test cases for the following pseudo-code to achieve full branch coverage:

function getDiscount(userType, total): if userType == "student":

if total > 1000:

return 20

else:

return 10

else:

return 5

Ans.

To achieve **100% branch coverage**, every true and false branch must execute.

TC ID	User Type	Total	Expected Output	Branch Covered
TC1	student	1200	20	Outer IF = True, Inner IF = True
TC2	student	800	10	Outer IF = True, Inner IF = False
TC3	regular	1500	5	Outer IF = False

Coverage Achieved: 100% Branch Coverage.

3. For this pseudo-code that checks payment method and amount, write test cases to achieve condition coverage:

function isPaymentAllowed(method, amount):

if (method == "UPI" or method == "Card") and amount <= 10000:

return true

else:

return false

Ans.

Conditions :-

- **C1:** method == "UPI"
- **C2:** method == "Card"
- **C3:** amount <= 10000

For **condition coverage**, each condition should evaluate to both **True** and **False** at least once.

TC ID	Method	Amount	C1	C2	C3	Expected Output
TC1	UPI	5000	True	False	True	True
TC2	Card	9000	False	True	True	True
TC3	Cash	5000	False	False	True	False
TC4	UPI	15000	True	False	False	False

Coverage Achieved:

- C1 → True & False ✓
- C2 → True & False ✓
- C3 → True & False ✓

4. Given the pseudo-code below for a simple login check, write white-box test cases to cover all unique execution paths (path coverage):

```
function login(username, password):
```

```
  if username == "admin":
```

```
    if password == "pass123":
```

```
      return "Login Success"
```

```
    else:
```

```
      return "Wrong Password"
```

```
  else:
```

```
    return "User Not Found"
```

Ans.

Unique Execution Paths :-

Path 1: Username = admin → Password correct → Login Success

Path 2: Username = admin → Password incorrect → Wrong Password

Path 3: Username ≠ admin → User Not Found

Test Cases :-

TC ID	Username	Password	Expected Output	Path Covered
TC1	admin	pass123	Login Success	Path 1
TC2	admin	abc123	Wrong Password	Path 2
TC3	user1	pass123	User Not Found	Path 3

Coverage Achieved: 100% Path Coverage (all unique execution paths executed).

